Hand Gesture Control - User Guide

Control your Windows PC with your hand and a webcam. Everything runs on your PC - no video is saved or sent anywhere.

1. Start the app

1. Close other apps that use the camera (Zoom, Teams, Camera app).
2. Double-click the **Hand Gesture Control** icon on your Desktop (blue circle with a white hand).
   No icon? Open **C:\Users\SUHAIL KHAN\hand-gesture-control** and double-click **start.bat**.
3. Two windows open: a black **console** (shows a log of actions) and the **camera preview**. Keep both open.
4. The app starts **PAUSED**. Hold an **open palm** 🖐 **for 1 second** - it turns **ACTIVE**.

2. Where to put your hand

- Sit **50-80 cm** from the camera, palm facing it.
- Keep your hand inside the **yellow rectangle** in the preview. That rectangle = your whole screen, so small moves are enough.
- Good light on your hand helps a lot.
- Your **right hand** controls; your **left hand** takes over if the right one is out of view. The hand in control is drawn **blue**, an ignored hand **grey**.

3. PAUSED and ACTIVE - what they mean

Mode	What happens
PAUSED	The app sees your hand but **ignores it**. Cursor does not move, nothing is clicked.
ACTIVE	Your gestures **control the PC**.

It is a safety switch: pause it to scratch your head, drink water, or talk with your hands without clicking things.

The mode is shown in the **top-left corner** of the preview.

4. Switch between them

To	Gesture	Keyboard
PAUSE	Fist ✊ - hold **1 s**	`Ctrl` + `Alt` + `P`
RESUME	Open palm 🖐 - hold **1 s**	`Ctrl` + `Alt` + `P`

While you hold, a **yellow progress bar** fills on the right. Open your hand before it is full to cancel.

5. Why did it pause by itself?

- It **always starts paused**.
- Your hand rested in a **loose fist** for over a second - that counts as the pause gesture.
- You pressed `Ctrl` + `Alt` + `P`.

> Not using it for a moment? Just move your hand **out of the camera view**. No hand = nothing happens.

6. Stop the app

- `Ctrl` + `Alt` + `Q` (works anywhere)
- or `Esc` while the preview window is selected
- or close the preview window with its **X**

Emergency stop: push your **real mouse** into any screen corner - the app stops immediately.

7. Helpers

- Press `H` in the preview: list of all gestures.
- **start-dry-run.bat**: practice mode, shows gestures but clicks nothing.
- **calibrate.bat**: measures your hand so gestures fit you better.

Step by step: everyday tasks

Make sure the preview shows **ACTIVE** (green) before you start.

📁 Open a folder

1. **Point** ☝ (index finger up, others folded). Move your hand slowly - the cursor follows your fingertip.
2. Stop with the cursor **on the folder icon**. Keep your hand still.
3. **Two quick pinches** 👏👏: touch thumb + index tips, open, touch, open - as fast as a mouse double click.
4. The folder opens. The preview shows **LEFT CLICK** then **DOUBLE CLICK**.

> The cursor **freezes** while your fingers close, so it will not slide off the icon.
> **Other way:** one pinch to select the folder, then **thumbs up** 👍 **(hold 0.7 s)** = Enter = open.

Only one click happened? Pinch faster. **It said DRAG?** You held the pinch too long - make it a quick tap.

🗂️ Drag a file to another folder

1. **Point** ☝ the cursor onto the file.
2. **Pinch and keep holding** 👏 for about half a second - the preview shows **DRAG**.
3. Still pinching, move your hand - the file moves with the cursor.
4. Over the target folder, **open your fingers** - the file is dropped (**DROP**).

> If your hand leaves the camera while dragging, the file is let go automatically.

↕️ Scroll up and down (folder, web page, document)

1. **Point** ☝ the cursor onto the **area you want to scroll** (the file list, the web page). Windows scrolls whatever is under the cursor.
2. Make the **V sign** ✌ (index + middle up) and **hold still a moment** - this height becomes your **starting point**.
3. Move your hand up or down:

You do	Result
Hand **a little up** from the start point	Scrolls **up** slowly
Hand **further up**	Scrolls **up faster**
Hand **down** from the start point	Scrolls **down** (further = faster)
Hand **back to the start point**	Scrolling **stops**
Change gesture (e.g. back to ☝)	Scrolling **stops** completely

> It works like a **joystick**: hold your hand above the start point and it keeps scrolling. Tiny shakes near the start point do nothing.

📋 Right-click menu

1. **Point** ☝ at the item (file, desktop, text).
2. Touch **thumb + middle finger** tips 👌 - the menu opens.
3. Point at a menu entry and **pinch once** 👏 to choose it.

📌 Example: open Documents and scroll down

🖐 hold 1 s → ACTIVE → ☝ cursor on **Documents** → 👏👏 opens → ☝ cursor on the file list → ✌ hold, then lower hand = scroll down → raise back = stop → ✊ hold 1 s = PAUSED

More things you can do

Zoom, 3D, music, swipes and shortcuts - plus quick fixes.

🌍 Google Earth Pro / maps / 3D

- **Zoom:** three fingers 🖐, move hand up = zoom in, down = zoom out.
- **Zoom like VR:** pinch with **both hands** 🙌, pull apart = in, push together = out.
- **Move the map:** pinch & hold 👌 and move (drags the globe).
- **Tilt / rotate:** four fingers 🤚 (thumb folded), move your hand.

> The app notices Google Earth and Blender automatically and uses the right zoom for them.

🔎 Zoom in a browser, Word, photos

Three fingers 🖐 up/down = **Ctrl + mouse wheel** zoom. Put the cursor over the page first.

🎵 Music, video & volume

- **Volume:** call-me sign 🤙 (thumb + pinky out), move hand up = louder, down = quieter.
- **Play / Pause:** rock sign 🤘 (index + pinky up), hold 0.6 s.

👋 Swipes (open palm, quick flick)

Swipe	Normally	In a browser
←	Switch app	Back
→	Switch app	Forward
↑	Task view	Task view
↓	Show desktop	Show desktop

In **PowerPoint**: ← / → = previous / next slide. A still palm never swipes.

⌨️ Keys & shortcuts (hold the gesture)

- Thumbs up 👍 0.7 s = **Enter**
- Thumbs down 👎 0.8 s = **on-screen keyboard**
- Pinky only 0.8 s = **screenshot** (Win+Shift+S)

🛠️ Something not working?

Problem	Try this
"Could not open camera"	Close Zoom/Teams/Camera app; check Windows camera privacy settings.
Nothing happens	Is it **PAUSED**? Hold open palm 1 s. Is your hand inside the yellow rectangle?
Pauses by itself	Do not rest a closed hand in view; move it away instead.
Clicks become drags	Pinch quicker (a short tap).
Gestures too easy / too hard	Run **calibrate.bat**.
Cursor shaky	Better light, hand closer, keep it inside the rectangle.
Scroll too fast / wrong way	Change `scroll.speed` / `scroll.invert` in config.json.
Wrong hand controls	Change `tracking.hand` in config.json.

🔒 Privacy

The camera image is processed on your PC and thrown away right after. Nothing is recorded, saved or uploaded.

Hand Gesture Control - Cheat Sheet

Control your mouse and keyboard with your hand through the webcam. 100% local, nothing is recorded.

Starts **PAUSED** - hold an open palm to begin

MOUSE

☝️ **Point**
Index finger up, others down. Fingertip moves the cursor.
Move cursor

👌 **Pinch (tap)**
Quickly touch thumb + index tips, then open.
Left click

👌👌 **Pinch twice**
Two quick pinches in a row.
Double click

👌⏳ **Pinch & hold**
Hold thumb + index > 0.4 s, move, open to drop.
Drag & drop

👌 **Middle pinch**
Touch thumb + **middle** fingertip.
Right click

✌️ **V sign**
Index + middle up. Move hand up/down.
Scroll

🖖 **Three fingers**
Index + middle + ring up. Move hand up/down.
Zoom in / out

🤚 **Four fingers**
All fingers up, thumb folded. Move hand around.
Rotate 3D view

🙌 **Two-hand zoom**
Pinch with **both** hands, pull apart / push.
Zoom in / out

🤙 **Call-me sign**
Thumb + pinky out. Move hand up/down.
Volume up / down

✊ **Fist**
Close your whole hand and hold.
Pause control hold 1 s

🖐️ **Open palm**
All 5 fingers spread, hold still.
Resume control hold 1 s

KEYS, MEDIA & SHORTCUTS

👍 **Thumbs up**
Fist with thumb pointing up.
Enter hold 0.7 s

🤘 **Rock sign**
Index + pinky up, thumb folded.
Play / Pause hold 0.6 s

👎 **Thumbs down**
Fist, thumb pointing down.
On-screen keys hold 0.8 s

🤙 **Pinky only**
Only the little finger up, thumb folded.
Screenshot hold 0.8 s

SWIPES - OPEN PALM, QUICK FLICK (A STILL PALM NEVER SWIPES)

🖐️⬅️ **Swipe left**
Switch app

🖐️➡️ **Swipe right**
Switch app

🖐️⬆️ **Swipe up**
Task view

🖐️⬇️ **Swipe down**
Show desktop

Apps, hotkeys & tips

Some gestures change automatically depending on the app in front (shown as "App profile" in the preview).

PER-APP PROFILES

App	What changes
Chrome, Edge, Firefox, Brave, Opera	Swipe ← = **Back**, swipe → = **Forward**
PowerPoint	Swipe ← = **previous slide**, swipe → = **next slide**
Google Earth Pro	3 fingers & two-hand pinch = **zoom** (mouse wheel) 4 fingers = **tilt / rotate** view · Swipe ← → = move the view · Pinch & hold = drag the globe
Blender	3 fingers & two-hand pinch = **zoom** · 4 fingers = **orbit**
Everything else	Default gestures from page 1 (3 fingers = Ctrl+wheel zoom)

START & STOP

What	How
Start (real control)	Double-click **start.bat** - starts PAUSED, hold an open palm 🖐
Test only (no clicks)	Double-click **start-dry-run.bat**
Fit to your hand	Double-click **calibrate.bat** and follow the 4 steps
Quit instantly	`Ctrl` + `Alt` + `Q` or `Esc` in the preview
Pause / resume	`Ctrl` + `Alt` + `P` or fist ✊ / open palm 🖐
Gesture list	`H` in the preview window
Emergency stop	Push the **real mouse** into any screen corner

TIPS FOR BEST RESULTS

- Sit **50-80 cm** from the camera, palm facing it.
- Keep your hand inside the **yellow rectangle** - it covers the whole screen, no need to reach the edges.
- Good, even **light** on your hand; avoid a bright window behind you.
- A plain background helps tracking.
- **Right hand** is in control; your **left hand** takes over when the right one is out of view. The hand in control is drawn **blue**, an ignored hand **grey**.
- Finger names on screen turn **green** when that finger counts as "up".
- Make gestures clearly and hold them still for a moment - each gesture needs a few steady frames.

READING THE PREVIEW WINDOW

- **ACTIVE / PAUSED / DRY RUN** - top left
- **Gesture:** what it sees now · **FPS** top right (25+ is good)
- **Yellow text** - the action just done (e.g. LEFT CLICK)
- **Progress bar** - a hold gesture charging up
- **Bottom lines** - fingers up, pinch distance, which hand

SOMETHING FEELS OFF?

Clicks become drags → pinch quicker, or raise `timing.drag_hold_s`.
Gestures trigger too easily / too hard → run **calibrate.bat**.
Cursor shaky / slow → tune `smoothing` in config.json.
Turn a gesture off → set it to `null` in `gesture_actions`.